Mediation Analysis Example

Davood Tofighi, Ph.D.

Scenario Setup

Research Question: What is the effect of a job training program (A) on income (Y), and how much of this effect is mediated through skill level (M)?

- **Treatment** (A): Job training program ($A = 0$: no training, $A = 1$: training)
- **Mediator** (M): Skill level (continuous scale 0–100)
- **Outcome** (Y): Annual income (in thousands)
- **Sample size**: $n = 1{,}000$ individuals

Generating Models

The following are **generating models for observed outcomes**, which under consistency also determine the **potential outcomes** M^a and $Y^{a,m}$.

Mediator Model (for M, a Function of A)

$$M = \alpha_0 + \alpha_1 A + \varepsilon_M$$

- $\alpha_0 = 60$: baseline skill level
- $\alpha_1 = 15$: effect of training on skills

From this model:

- $M^1 = \alpha_0 + \alpha_1 = 75$
- $M^0 = \alpha_0 = 60$

Outcome Model (for Y, as a Function of A, M, and $A \cdot M$)

$$Y = \beta_0 + \beta_1 A + \beta_2 M + \beta_3 AM + \varepsilon_Y$$

- $\beta_0 = 30$: baseline income
- $\beta_1 = 8$: direct effect of training
- $\beta_2 = 0.4$: effect of skills on income
- $\beta_3 = 0.1$: interaction effect between training and skills

Comprehensive Calculation Table

Component	Formula	Calculation	Value
COUNTERFACTUAL MEDIATOR VALUES			
M^1	$\alpha_0 + \alpha_1 \cdot 1$	$60 + 15$	**75**
M^0	$\alpha_0 + \alpha_1 \cdot 0$	$60 + 0$	**60**
COUNTERFACTUAL OUTCOMES			
Y^{1,M^1}	$\beta_0 + \beta_1 + \beta_2 M^1 + \beta_3 \cdot 1 \cdot M^1$	$30 + 8 + 0.4 \cdot 75 + 0.1 \cdot 75$	**75.5**
Y^{0,M^1}	$\beta_0 + 0 + \beta_2 M^1 + 0 \cdot \beta_3 \cdot M^1$	$30 + 0 + 0.4 \cdot 75$	**60.0**
Y^{1,M^0}	$\beta_0 + \beta_1 + \beta_2 M^0 + \beta_3 \cdot 1 \cdot M^0$	$30 + 8 + 0.4 \cdot 60 + 0.1 \cdot 60$	**68.0**
Y^{0,M^0}	$\beta_0 + 0 + \beta_2 M^0 + 0 \cdot \beta_3 \cdot M^0$	$30 + 0 + 0.4 \cdot 60$	**54.0**
CAUSAL EFFECTS			
Natural Direct Effect (NDE)	$Y^{1,M^0} - Y^{0,M^0}$	$68.0 - 54.0$	**14.0**
Natural Indirect Effect (NIE)	$Y^{1,M^1} - Y^{1,M^0}$	$75.5 - 68.0$	**7.5**
Total Effect (TE)	$Y^{1,M^1} - Y^{0,M^0}$	$75.5 - 54.0$	**21.5**
Alternative NIE	$Y^{0,M^1} - Y^{0,M^0}$	$60.0 - 54.0$	**6.0**
VERIFICATION			
NDE + NIE = TE?	$14.0 + 7.5$		**21.5** ✓
PROPORTIONS			
% Mediated	$\frac{\text{NIE}}{\text{TE}} \times 100$	$\frac{7.5}{21.5} \times 100$	**34.9%**
% Direct	$\frac{\text{NDE}}{\text{TE}} \times 100$	$\frac{14.0}{21.5} \times 100$	**65.1%**

Interpretation

Key Findings

1. **Total Effect**: The job training program increases income by $21,500 on average.
2. **Natural Direct Effect**: $14,000 (65.1%) of the effect operates through pathways other than skill improvement.
3. **Natural Indirect Effect**: $7,500 (34.9%) of the effect is mediated via increased skill levels.
4. **Mediation**: Roughly one-third of the program's benefit comes from skill enhancement.

Counterfactual Interpretation

- $Y^{1,M^1} = 75.5$: Income if treated and skill develops as it would under treatment.
- $Y^{1,M^0} = 68.0$: Income if treated but with skill level that would occur under no treatment.
- $Y^{0,M^1} = 60.0$: Income if untreated but skill level is as if treated.
- $Y^{0,M^0} = 54.0$: Income if untreated and skill level is as under no treatment.

Policy Implications

- The program has **direct effects** beyond skills (e.g., networking, confidence, signaling).
- The **mediator (skills)** contributes significantly but is not the dominant pathway.
- Interventions should support both **skill acquisition** and **broader contextual benefits**.

Technical Notes

- These are **structural models** for observed variables that also imply counterfactuals under **consistency and ignorability**.
- The interaction term $\beta_3 = 0.1$ introduces **non-additivity** across treatment levels.
- The alternative NIE (e.g., $Y^{0,M^1} - Y^{0,M^0}$) differs due to this interaction.
- **Inference procedures** (e.g., bootstrapping or delta method) and **sensitivity analyses** are required in practice.

Worked Example: Calculating NDE, NIE, and TE for Three Individuals

In this example, we use a structural equation model to simulate data for three individuals and calculate the **Natural Direct Effect (NDE)**, **Natural Indirect Effect (NIE)**, and **Total Effect (TE)** of a binary treatment A on an outcome Y, mediated through variable M.

Model Setup

We assume the following structural equations:

Mediator Model

$$M = \alpha_0 + \alpha_1 A + \varepsilon_M$$

- $\alpha_0 = 60, \alpha_1 = 15$

Outcome Model

$$Y = \beta_0 + \beta_1 A + \beta_2 M + \beta_3 A \cdot M + \varepsilon_Y$$

- $\beta_0 = 30, \beta_1 = 8, \beta_2 = 0.4, \beta_3 = 0.1$

Simulated Individual-Level Data

ID	A	ε_M	ε_Y	M^1	M^0	Y^{1,M^1}	Y^{1,M^0}	Y^{0,M^1}	Y^{0,M^0}	NDE	NIE	TE
1	1	0	0	75	60	75.5	68.0	60.0	54.0	14.0	7.5	21.5
2	0	-3	2	72	57	76.0	68.5	60.8	54.8	13.7	7.5	21.2
3	1	5	-1	80	65	77.0	69.5	61.0	55.0	14.5	7.5	22.0

Step-by-Step: Individual 1

Step 1: Compute Potential Mediators

- $M^1 = 60 + 15 \cdot 1 + 0 = 75$
- $M^0 = 60 + 15 \cdot 0 + 0 = 60$

Step 2: Compute Potential Outcomes

- $Y^{1,M^1} = 30 + 8 + 0.4 \cdot 75 + 0.1 \cdot 75 + 0 = 75.5$
- $Y^{1,M^0} = 30 + 8 + 0.4 \cdot 60 + 0.1 \cdot 60 + 0 = 68.0$
- $Y^{0,M^1} = 30 + 0 + 0.4 \cdot 75 + 0 + 0 = 60.0$
- $Y^{0,M^0} = 30 + 0 + 0.4 \cdot 60 + 0 + 0 = 54.0$

Step 3: Calculate Effects

- **NDE** = $Y^{1,M^0} - Y^{0,M^0} = 68.0 - 54.0 = 14.0$
- **NIE** = $Y^{1,M^1} - Y^{1,M^0} = 75.5 - 68.0 = 7.5$
- **TE** = $Y^{1,M^1} - Y^{0,M^0} = 75.5 - 54.0 = 21.5$

Interpretation

- The **Total Effect** for individual 1 is **21.5**, which is decomposed into:
  - **Natural Direct Effect**: 14.0 (not mediated by skill)
  - **Natural Indirect Effect**: 7.5 (mediated through skill development)
- The proportion mediated is:

$$\frac{\text{NIE}}{\text{TE}} = \frac{7.5}{21.5} \approx 34.9\%$$

Observations Across Individuals

- All individuals show similar **NIE** due to the same change in mediator under treatment ($\Delta M = 15$).
- Variation in **TE** is driven by individual residuals ε_M and ε_Y.
- Slight differences in **NDE** and **TE** arise from heterogeneity introduced through ε_Y and the interaction term $\beta_3 AM$.

Summary

This example illustrates how to use structural models to simulate potential outcomes and decompose the total causal effect into direct and indirect components for **each unit**. These unit-level calculations help reveal individual heterogeneity in response to interventions—crucial in personalized medicine, education, or employment policy evaluation.